KEY INSIGHTS — WEATHER DATASET ANALYSIS (2008–2017)

- **Strong seasonality confirmed (Southern Hemisphere pattern):** temperatures peak in Dec–Feb and drop to their lowest in Jun–Aug, consistent across all 10 years.
- **Winter is the wettest season:** June alone accounts for the highest monthly rainfall total, followed by the March–April period; summer months are comparatively dry.
- **Humidity is the strongest rainfall signal:** high Humidity3pm values consistently precede both same-day rainfall and a "RainTomorrow = Yes" outcome; low-humidity days show almost no rain.
- **Cloud cover and sunshine are strongly inversely related** ($r \approx -0.67$), as expected; humidity in the afternoon is also negatively linked to the daily temperature range ($r \approx -0.63$) — humid days have smaller swings between min and max temperature.
- **99 days (3.03%) flagged as anomalous** via Isolation Forest, falling into two distinct types: extreme-rainfall days (high humidity, below-average pressure) and extreme heat/dry days (very low humidity, low pressure, no rain).
- **Anomalies cluster around season transitions** (Sep–Oct–Dec), suggesting less stable weather systems during the shift from spring to summer.
- **NW and S are the strongest wind directions**, relevant for identifying higher storm-risk conditions.
- **Data quality was high:** no missing values or duplicate records; all logical consistency checks (e.g. MinTemp ≤ MaxTemp, humidity within 0–100%) passed with zero violations.
- **Limitation:** 162 days (4.72%) are missing from the date range and were left undocumented rather than interpolated, to avoid introducing fabricated values. The dataset also contains no city/country field, so cross-location comparison (mentioned in the original task) was not possible — all findings are for a single station.